

Digital Signal Processing - Lecture 1

Linear Time-Invariant (LTI) systems

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Overview

About the course

- 6 lectures
- Emphasis on applying DSP using Python
- 50% of Introduction to Smart Systems Engineering

Some books

Digital Signal Processing First

James H. McClellan, Ronald W. Schafer, Mark A. Yoder

ISBN 9781292113869

Source: https://api.pageplace.de/preview/DT0400.9781292113876_A27168748/preview-9781292113876_A27168748.pdf

Think DSP

Allen B. Downey

ISBN 9781491938515

Source: <https://greenteapress.com/thinkdsp/thinkdsp.pdf>

Engineering Mathematics

ISBN 978-0273719779

Source: <https://www.amazon.com/Engineering-Mathematics-4th-edn-Communications/dp/0273719777>

Where is DSP used?

Where do you encounter digital signal processing?

Applications of DSP

Applications [\[edit \]](#)

General application areas for DSP include

- Audio signal processing
- Audio data compression e.g. MP3
- Video data compression
- Computer graphics
- Digital image processing
- Photo manipulation
- Speech processing
- Speech recognition
- Data transmission
- Radar
- Sonar
- Financial signal processing
- Economic forecasting
- Seismology
- Biomedicine
- Weather forecasting

Specific examples include [speech coding](#) and transmission in digital [mobile phones](#), [room correction](#) of sound in [hi-fi](#) and [sound reinforcement](#) applications, analysis and control of [industrial processes](#), [medical imaging](#) such as [CAT scans](#) and [MRI](#), [audio crossovers](#) and [equalization](#), [digital synthesizers](#), and audio [effects units](#).^[22]

Source: https://en.wikipedia.org/wiki/Digital_signal_processing

This lecture

01 Linear time-invariant systems

02 Discrete-time unit impulse

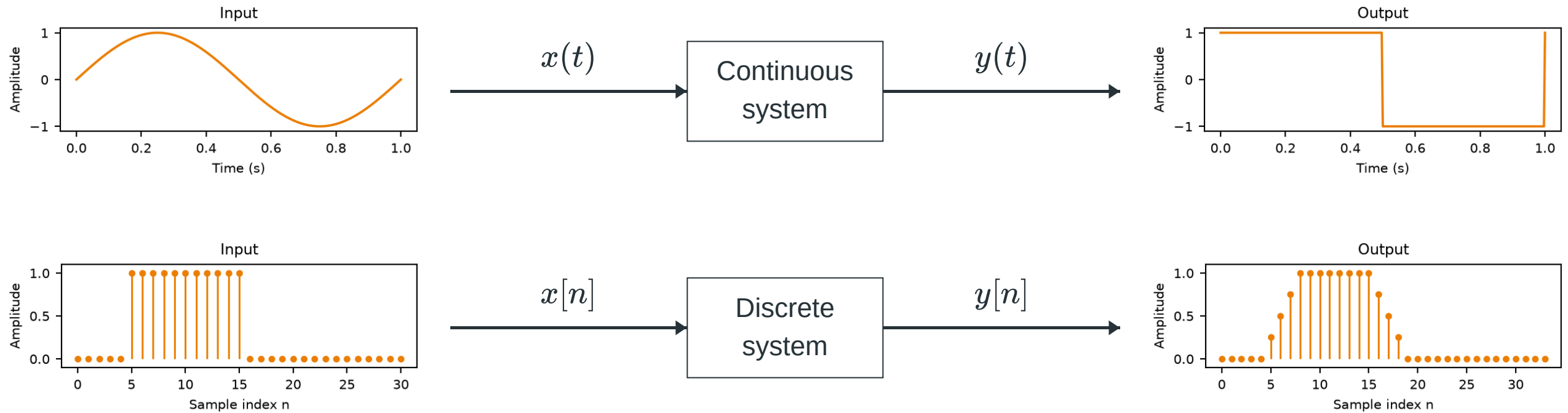
03 Discrete-time impulse response

04 Exercises

LTI systems

Linear time-invariant systems

System



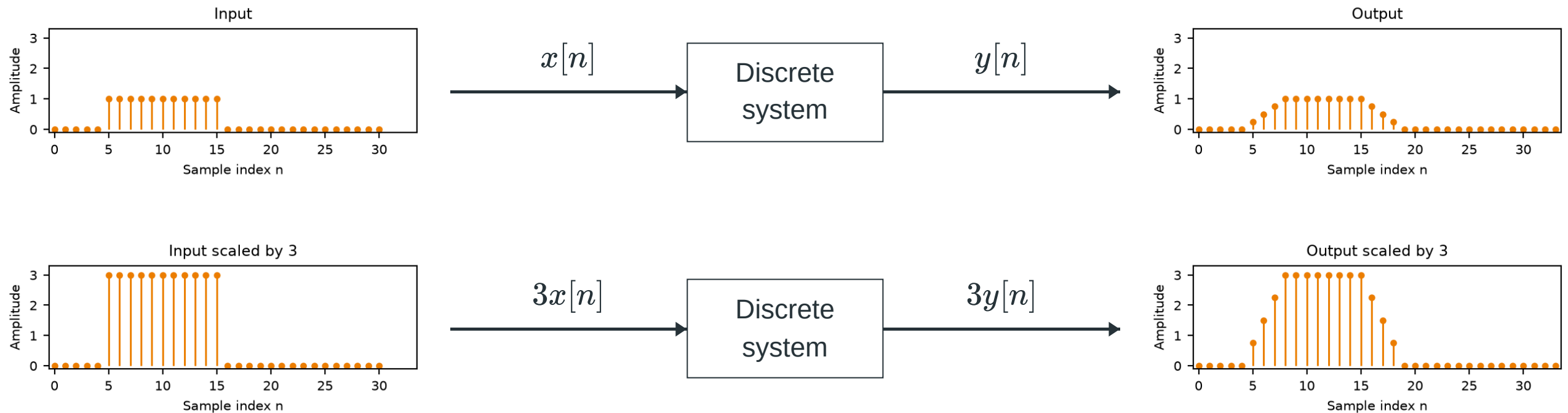
A system transforms an input signal into an output signal.

Linear time-invariant system

- Linearity consists of **homogeneity** and **additivity**.
- **Shift invariance** is also called time invariance.

Homogeneity

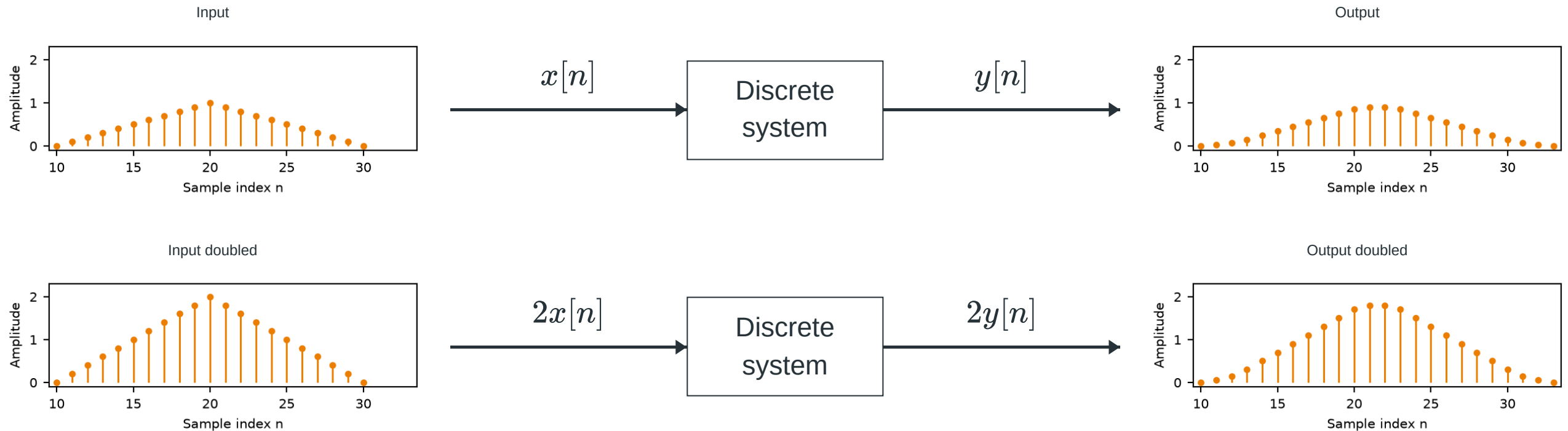
Both discrete systems represent the same system.



Multiplying the input by 3 also multiplies the output by 3.

Homogeneous? Example 1

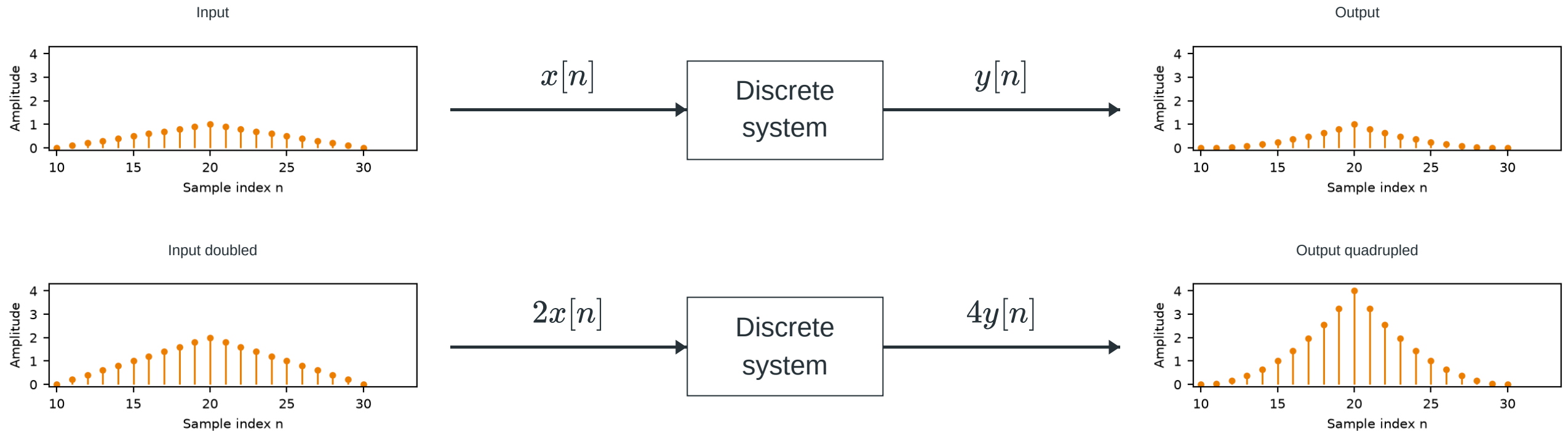
Both discrete systems represent the same system.



Homogeneous. Doubling the input doubles the output.

Homogeneous? Example 2

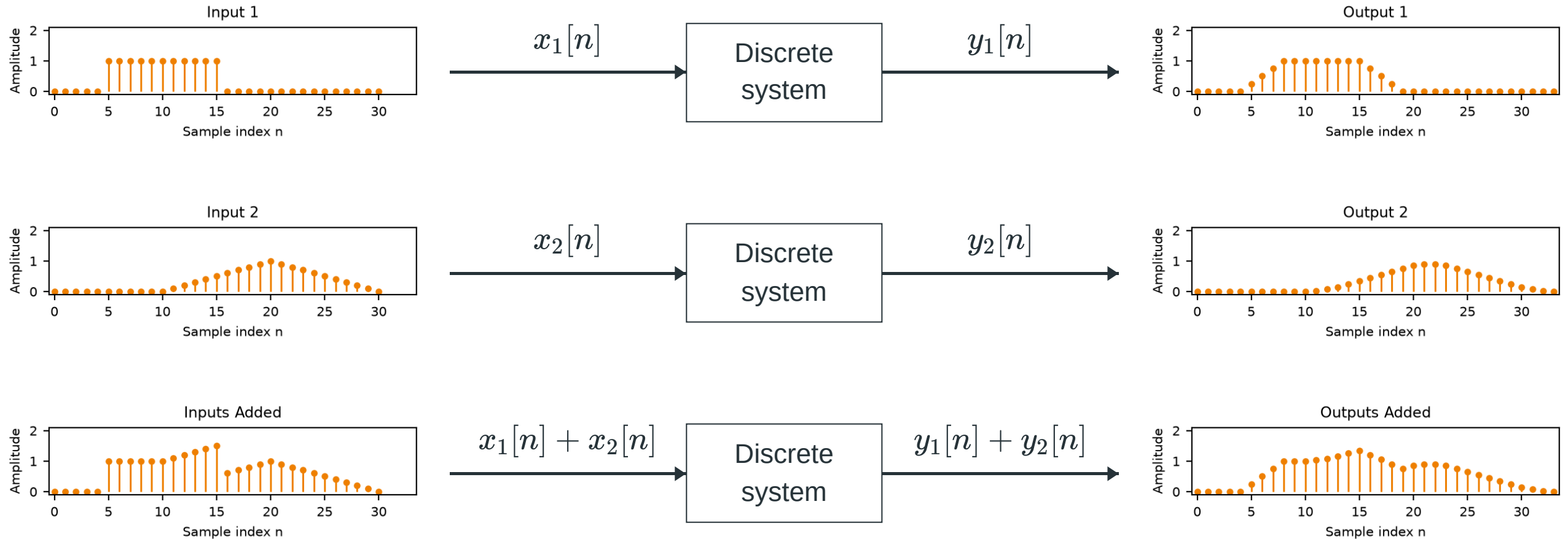
Both discrete systems represent the same system.



Not homogeneous. Doubling the input quadruples the output, rather than doubling it.

Additivity

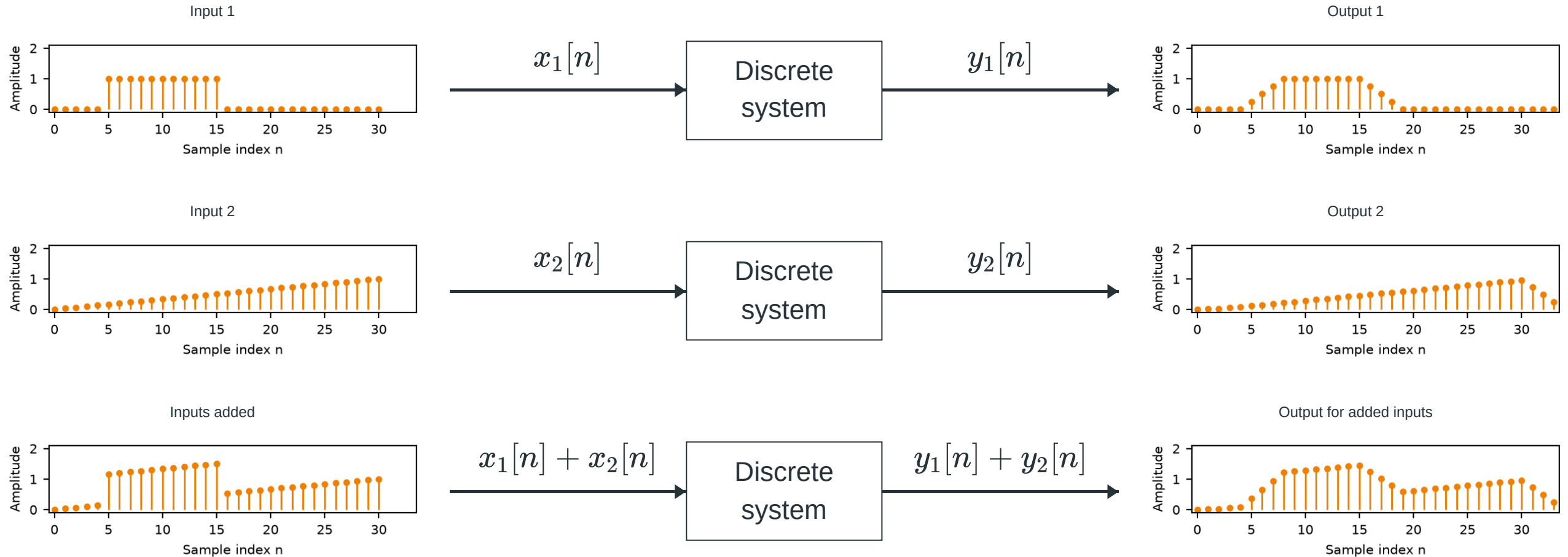
All three discrete systems represent the same system.



Adding the inputs before the system gives the same result as adding the outputs afterwards.

Additive?

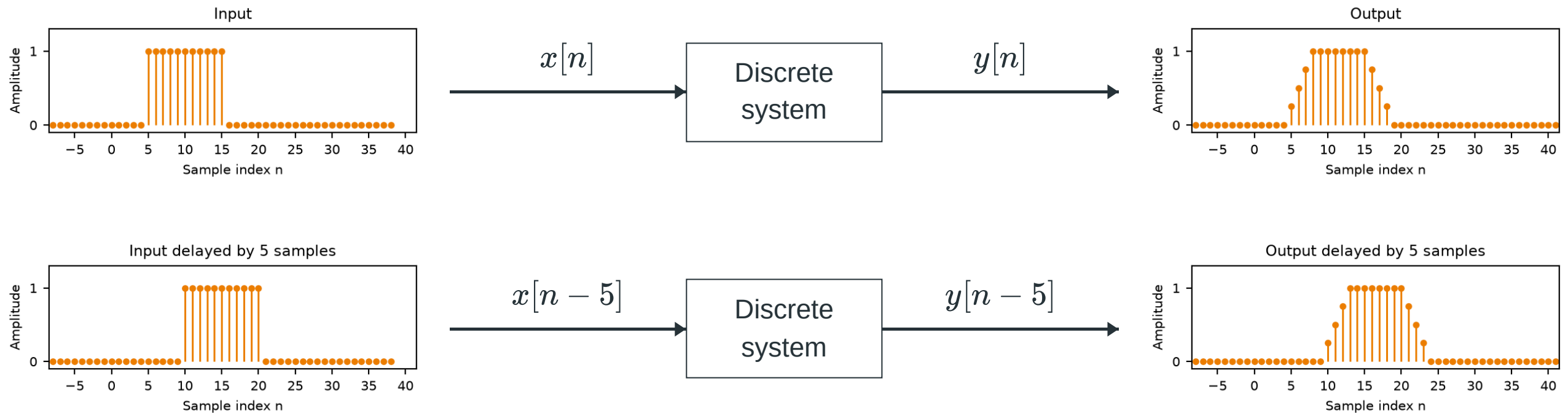
All three discrete systems represent the same system.



Additive. The output for the added inputs equals the sum of the two separate outputs.

Shift invariance

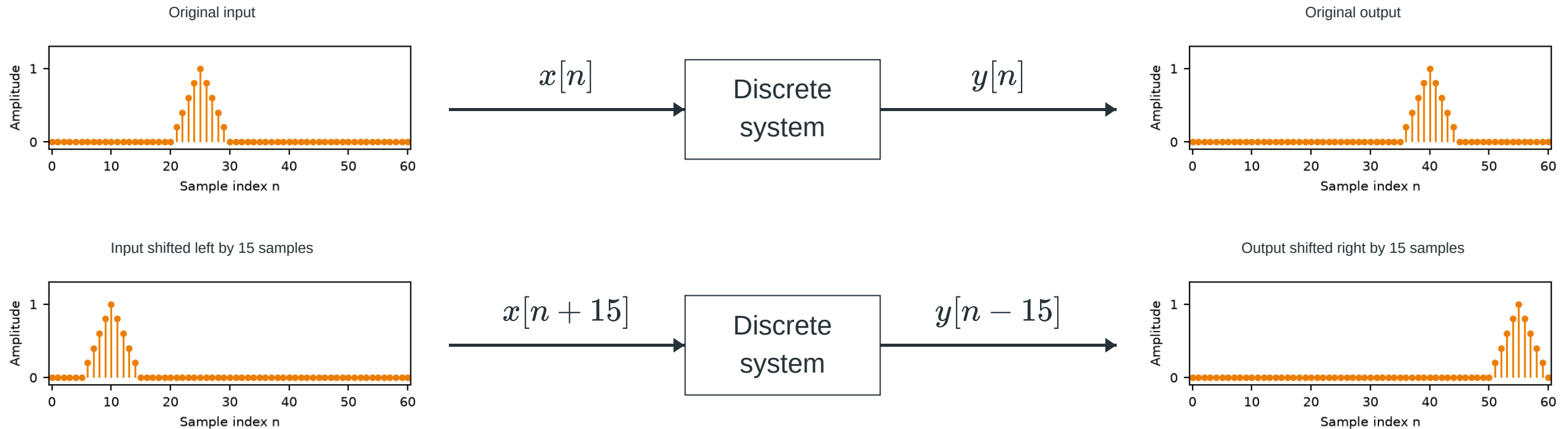
Both discrete systems represent the same system.



Shift invariant. Delaying the input by 5 samples delays the output by the same 5 samples.

Shift invariant?

Both discrete systems represent the same system.



Not shift invariant. The input moves left, but the actual output moves right.
In other words, when the input occurs earlier, the output occurs later.

LTI systems in three rules

A system that creates an output $y[n]$ for a given input $x[n]$ is LTI if it satisfies the properties of linearity and shift invariance. In these slides, $\rightarrow$ indicates the input-output relationship of the system.

Homogeneity

Scale input and output equally.

$$a x[n] \rightarrow a y[n]$$

Additivity

Add inputs and their outputs.

Given $x_1[n] \rightarrow y_1[n]$ and $x_2[n] \rightarrow y_2[n]$

$$x_1[n] + x_2[n] \rightarrow y_1[n] + y_2[n]$$

Shift invariance

Shift input and output equally. Also referred to as time invariance.

$$x[n - n_0] \rightarrow y[n - n_0]$$

Homogeneity + additivity = linearity

Linearity + shift invariance = LTI

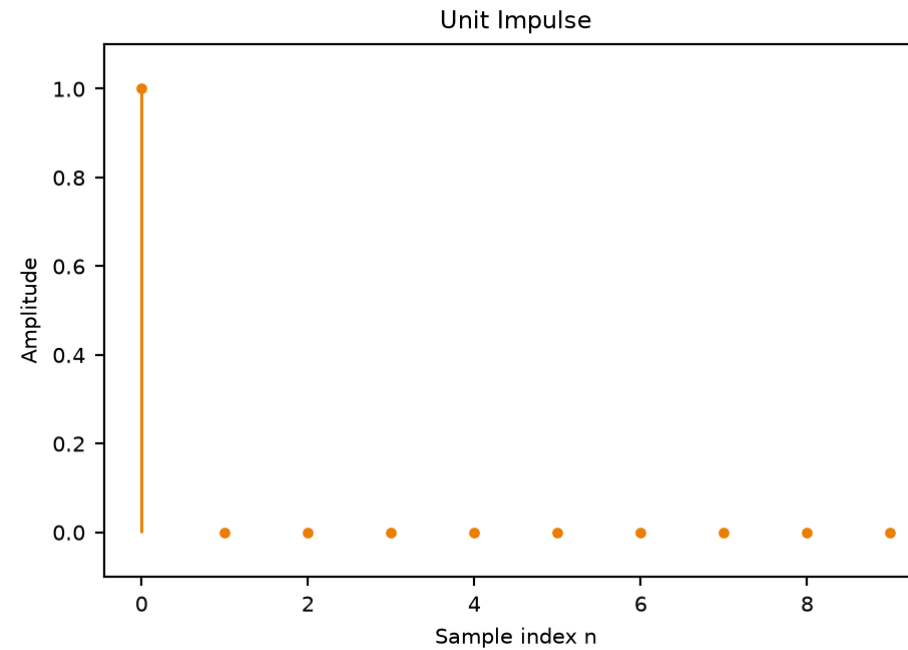
Questions?

Unit impulse

Discrete-time unit impulse

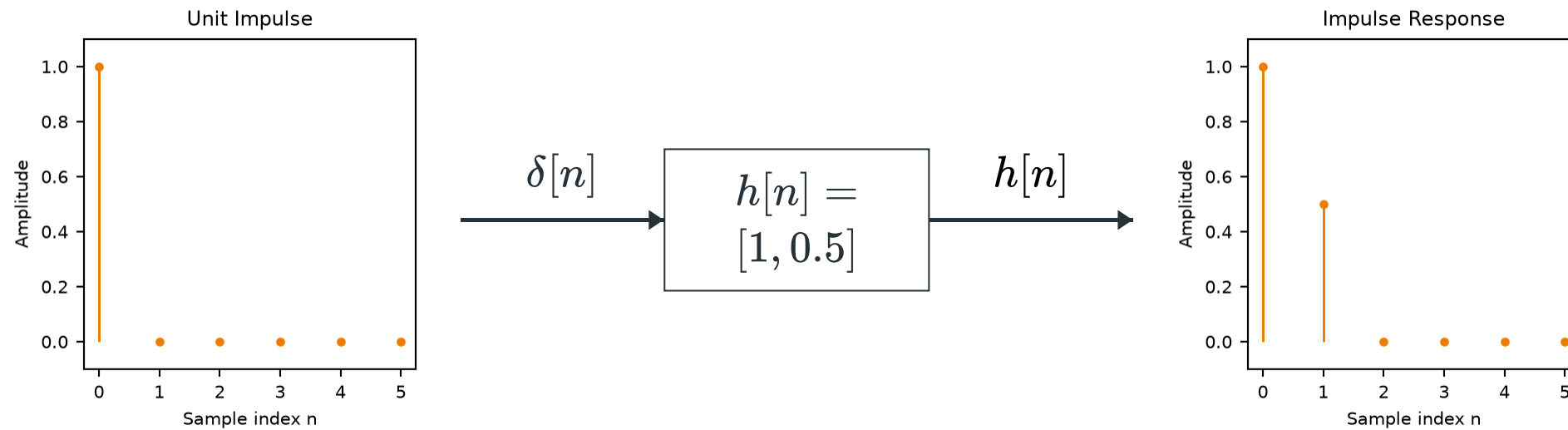
Discrete unit impulse

$$\delta[n] = \begin{cases} 1, & n = 0, \\ 0, & n \neq 0. \end{cases}$$



| Impulse response

Impulse response

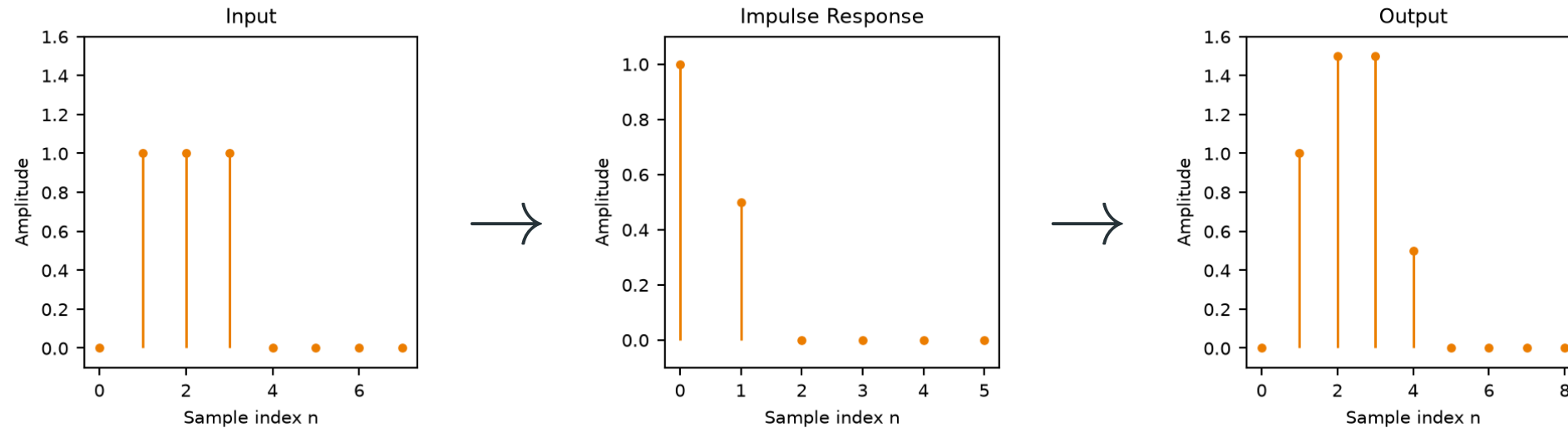


The output for a unit impulse is the system's impulse response.

Discrete-time impulse response

$$x[n] \longrightarrow h[n] \longrightarrow y[n]$$

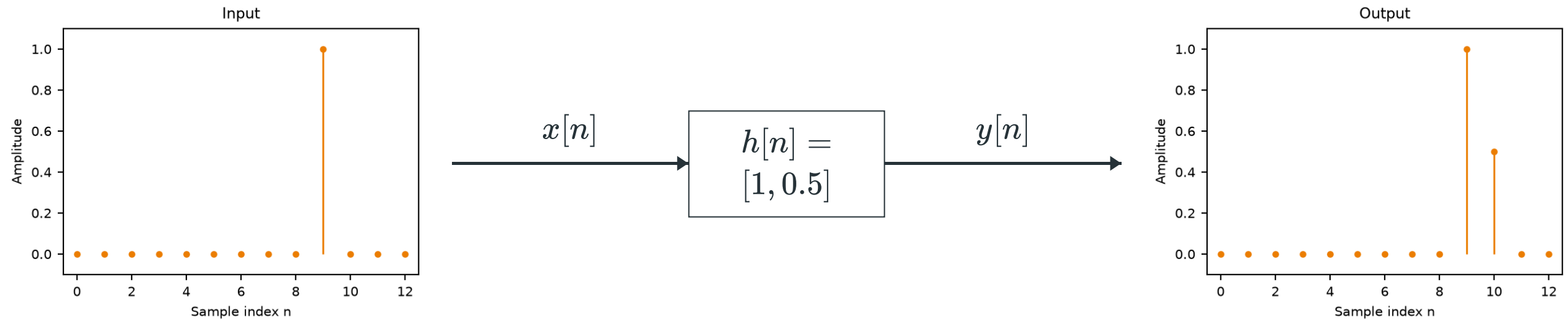
Discrete-time impulse response example



$$x[n] \longrightarrow h[n] \longrightarrow y[n]$$

For an LTI system, add scaled and shifted copies of $h[n]$ to find the output.

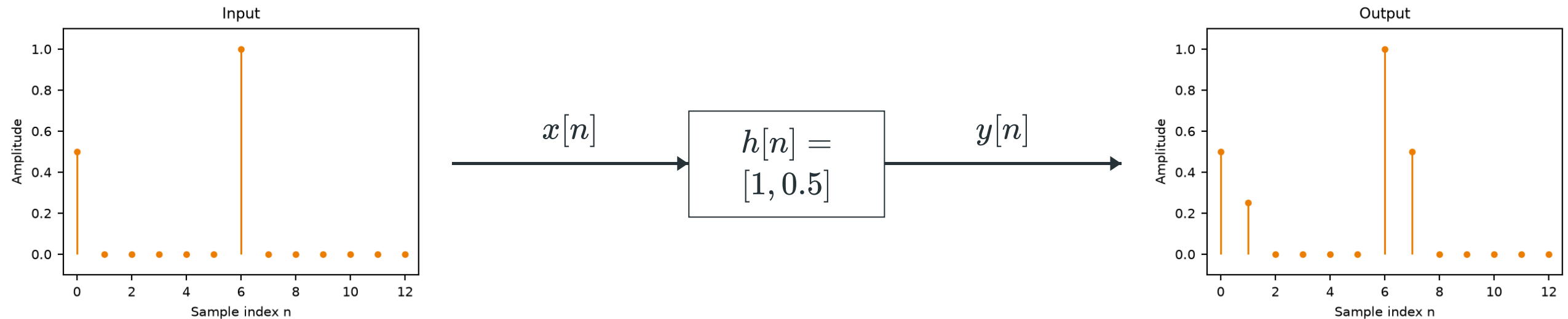
What input happened here? 1



$$x[n] = \delta[n - 9]$$

The output is $h[n]$ shifted 9 samples right, so the input is a unit impulse at $n = 9$.

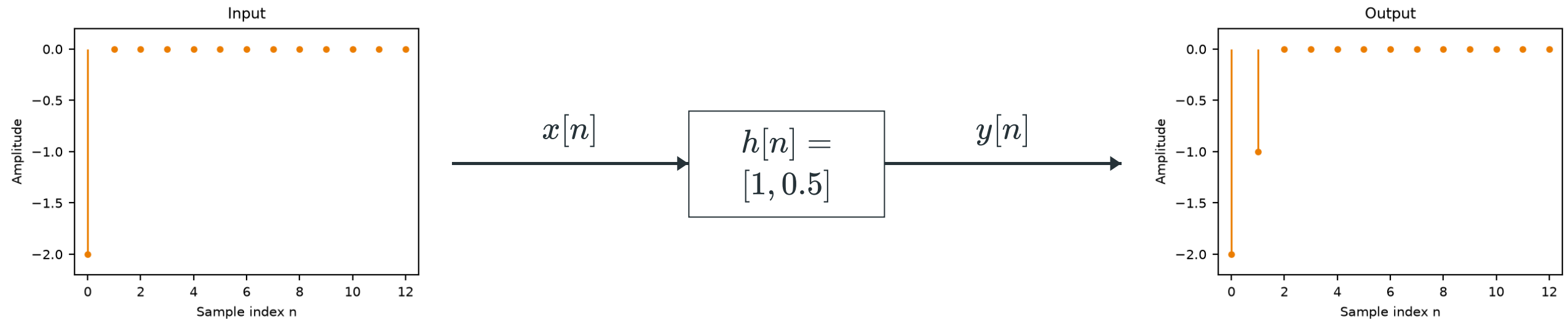
What input happened here? 2



$$x[n] = 0.5\delta[n] + \delta[n - 6]$$

The output combines half of $h[n]$ at $n = 0$ and a full copy at $n = 6$, so the input contains impulses of 0.5 and 1 at those indices.

What input happened here? 3



$$x[n] = -2\delta[n]$$

The output is $h[n]$ multiplied by -2 without a shift, so the input is an impulse of -2 at $n = 0$.

| Exercises

Do not forget the exercises!

`../exercises/DSP-Lecture-01-Exercises.ipynb`